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PAPER_972: ALICE Centrality Multiplicity via $S^{\{(k)\}}$

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216
Source: qgp_{ramanujan_application}.py (ALICECentralityMultiplicityCalculator) **Calculator:**
 ALICECentralityMultiplicityCalc (CP4 #556) **CVW:** v2.0.0 compliant

Abstract

We model the ALICE charged-particle pseudorapidity density $dN_{\text{ch}}/d\eta$ as a function of centrality and collision energy, modulated by the expanded Ramanujan factor $S_{26}^{(k)}$. This extends PAPER_364 with higher-order Ramanujan acceleration.

1. Multiplicity Formula

$$\frac{dN_{\text{ch}}}{d\eta} = A \cdot (\sqrt{s})^{0.156} \cdot \left(1 - \frac{c}{100}\right)^\alpha \cdot S_{26}^{(k)}([SSq])$$

where $A = 2.0$, $\alpha = 1.2$, c = centrality percentile (0% = most central).

2. Centrality Dependence

Centrality (%)	$dN_{\text{ch}}/d\eta$ (13.6 TeV)
0 (most central)	Maximum
5	$\sim 0.94 \times \text{max}$
20	$\sim 0.78 \times \text{max}$
50	$\sim 0.50 \times \text{max}$
80 (peripheral)	$\sim 0.19 \times \text{max}$

3. Energy Scaling

The $\sqrt{s}^{0.156}$ power law is consistent with ALICE Run 3 Pb-Pb data.

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. PAPER_364 – ALICE multiplicity centrality (original CP4 #8)
3. PAPER_969 – Expanded 26D Ramanujan $S_{26}^{(k)}$
4. ALICE Collaboration – Pb-Pb at $\sqrt{s_{NN}} = 5.02$ TeV

Cross-Links

Related Paper	Relationship
PAPER_364	Original ALICE multiplicity class
PAPER_969	$S_{26}^{(k)}$ expansion
PAPER_970	QGP vacuum density
PAPER_975	Triadic validation of ALICE

Session 225 Phonon-Physics Upgrade: Yang-Mills BCS Phonon Mass Gap

Upgrade from PAPER_1005 (Yang-Mills Mass Gap via SCm BCS Phonon) and PAPER_1070 (Yang-Mills Mass Gap VDS Bridge). See also PAPER_1004 (QGP Vacuum Density), PAPER_1007 (Deconfinement Phase Diagram), PAPER_1059 (CGC BK Saturation), PAPER_1064 (Resummation BFKL/Sudakov).

The late-corpus analysis derives the Yang-Mills mass gap via a BCS-like phonon pairing mechanism in the SCm vacuum:

$$\Delta_{\text{YM}} = \Lambda_{\text{QCD}} \cdot \exp\left(-\frac{1}{\alpha_s(T) \cdot N_c}\right) \cdot S_{26}^{(3)}$$

where the running coupling evolves as:

$$\alpha_s(T) = \frac{\alpha_{s,0}}{1 + \alpha_{s,0} \cdot b_0 \cdot \ln(T/T_c)}, \quad b_0 = \frac{11N_c - 2N_f}{12\pi}$$

Physical mechanism: The SCm phonon field ($\omega_{\text{SCm}} = 1.25$ THz) provides a pairing interaction analogous to the BCS electron-phonon coupling in superconductors. Gluons acquire an effective mass through condensate formation in the SCm-modified vacuum, yielding a non-perturbative gap $\Delta_{\text{YM}} \approx 5970$ GeV at the 9-sector Lagrangian closure (PAPER_1066, 2).

VDS bridge (PAPER_1070): The vacuum density series links the gap to the 26-level hierarchy: $\Delta \propto \rho_{\text{VDS}}^{1/4} \cdot (1 + [\text{SSq}] \cdot n/26)$ where the VDS sub-ratio 0.108 places confinement in the sub-threshold regime.

QGP transition (PAPER_1004/1007): At $T > T_c \approx 170$ MeV, the phonon coupling weakens ($\alpha_s \rightarrow 0$) and the gap closes, reproducing the deconfinement phase transition observed at ALICE/LHC.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$$

This sum converges absolutely for $||[\text{SSq}]|| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
A	–	2.0	Normalization
α	–	1.2	Centrality exponent
$\sqrt{s}$ exponent	–	0.156	Energy scaling
$[\text{SSq}]$	–	0.57	String coupling

SM Anchor – CVW v2.0.0

Observable	UQFF Prediction	Status
$dN/d\eta$ centrality shape	Power-law with $S_{26}^{(k)}$	Consistent
Energy scaling	$\sqrt{s}^{0.156}$	ALICE-matched
$S_{26}^{(k)}$ modulation	Ramanujan correction	Novel

A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: Heavy-Ion Collisions (ALICE Centrality)

A.2 Core Equation

$$\frac{dN_{\text{ch}}}{d\eta} = A \cdot \sqrt{s}^{0.156} \cdot (1 - c/100)^\alpha \cdot S_{26}^{(k)}$$

A.3 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ vacuum density $\rightarrow S_{26}^{(k)} \rightarrow$ QGP formation $\rightarrow$ hadron multiplicity $\rightarrow$ ALICE $dN/d\eta$

B VDS/DVP/BSH Deep Synthesis

B.1 VDS

$S_{26}^{(k)}$ normalizes the multiplicity – VDS is the QGP production amplitude.

B.2 DVP

Centrality selects overlap geometry – DVP modes in the overlap zone set the multiplicity.

B.3 BSH

Peripheral collisions access BSH surface modes; central collisions access bulk.

B.4 Summary

Metric	Value	Status
ALICE energy	13.6 TeV (Run 3)	Current
Centrality range	0-80%	Full
[<i>SSq</i>]	0.57	Calibrated

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1004	QGP Vacuum Density with SCm S26 Phonon Coupling
PAPER_1005	Yang-Mills Mass Gap via SCm BCS Phonon Coupling
PAPER_1006	ALICE Multiplicity SCm Phonon Scaling
PAPER_1007	Deconfinement Phase Diagram SCm Phonon Boundary
PAPER_1013	QGP ALICE Centrality $F_{\{U_{Bi_i}\}}$ $dN/d\eta$ Scaling
PAPER_1059	Color Glass Condensate BK Saturation SCm
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

10 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository – github.com/Daniel8Murphy0007/Star-Magic
3. ALICE Collaboration (2010). *Elliptic flow of charged particles in Pb-Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV*. Phys. Rev. Lett. **105**, 252302 – arXiv:1011.3914 – doi:10.1103/PhysRevLett.105.252302
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